Understanding and caring for the solar system

Components, present needs and criteria for expansion.

Solar panels / 8 × 300 W

Modules convert sunlight into direct current. Eight 300 W modules total 2.4 kWp nominally. Actual production depends on sunlight, shade, temperature and orientation.

Inverter + MPPT / 3 kW · 48 V

The inverter converts DC to AC. The MPPT controller manages energy from the panels. Every expansion requires checking electrical limits and compatibility.

Energy storage / 2 × 5.12 kWh

Two 5.12 kWh batteries total 10.24 kWh nominally. Usable energy depends on losses, manufacturer limits and battery condition.

Protection and mounting / System care

Mounting, earthing, wiring and protection are selected through technical design. Electrical connections and work at height require qualified personnel.

Community services / Light · care · connection

Efficient lighting and connectivity can support learning and shared activities. Loads and schedules are measured and agreed; future uses require a new assessment.

CONCEPTUAL VISUALISATION

How the solar system will look



Conceptual image shared by coordination. Eight 300 W rooftop panels, with an inverter, protection and storage area. Layout and equipment are subject to technical confirmation; this does not depict a completed installation.

SERVICES AND CONTINUITY

Size the system from the needs

Present and future

Lighting, cameras and connectivity connect with safety, care and learning. The eight panels donated by Rafiqui total 2.4 kWp. Inverter selection considers growth, but each expansion of panels or batteries must check voltage, current, power, storage and compatibility limits against manufacturer specifications.

Do not confuse capacity with runtime

3 kW describes rated inverter power. 10.24 kWh describes nominal storage, not fully usable energy. A daily balance needs consumption, hours, simultaneity, startup loads, sunlight, shade and losses. Refrigeration, pumps and other future uses require reassessment.

Care plan

Agree visual inspection, alarm logs, technical response, warranties, costs and succession. Do not open protection devices or work on connections without appropriate competence and authority. The authorised video presents needs expressed by participants; it does not establish completed installation or sufficiency for every load mentioned.

Educational material; each installation requires its own design and review by qualified personnel.